

Shafwat Khan

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EDUCATION

University of Toronto - St. George

2024 – 2028 + PEY

Bachelor of Applied Science in Mechanical Engineering

Relevant Courses: Mechanical Design, Materials Science, Linear Algebra, Python Programming, Differential Equations, Numerical Methods, Statistics, Statics, Dynamics, Organizational Behaviour. (See Portfolio for Robotic Arm project)

Intended Minor: Advanced Manufacturing

SKILLS

Certifications: CSWA, CSWP, UofT Machine Shop

Technical: MATLAB, L^AT_EX, SolidWorks, Lantek Expert, CES EduPack, 3D Printing Slicing

Engineering: FEA, FEM, GD&T, DFM, Engineering Drawings, Circuit Building, Micro-Controllers, Tolerance

WORK EXPERIENCE

Engineering Support - Sable Metal Fabrication

May 2025 – Aug 2025

- Designed and modified custom sheet metal components in **SolidWorks**, applying **DFM** principles and realistic **manufacturing tolerances**; nested **30+ laser cutting** programs using **Lantek** to improve fabrication efficiency.
- Designed and installed a **welding fume exhaust system** by analyzing duct layouts and airflow requirements, selecting an appropriate **CFM-rated fan** to improve fume capture and extraction efficiency. (See Portfolio)
- Performed **thermal and structural simulations** on shop components (e.g., slat-wall shelving) to evaluate load capacity and safety factors, supporting improved workspace organization. (Slat-wall in portfolio)
- Produced and interpreted detailed **engineering drawings** using proper dimensioning and **GD&T** to support fabrication and quality control inspections.

LEADERSHIP & ENGINEERING EXPERIENCE

Student Engineer - UT Wind

Sep 2025 – Present

- Designed **3D-printable mechanical components** for the nacelle braking system and structural skeleton, applying **DFM** principles to ensure full printability (with limited bed size), reliable assembly, and structural integrity.
- Conducted **aerodynamic and thermal simulations** to analyze turbine blade behavior, airflow characteristics, and heat effects, supporting performance-driven design decisions.
- Performed **experimental testing** and **analytical force calculations** to quantify loads during turbine rotation; the results were used for the optimization of braking and support mechanisms.
- Improved overall **system efficiency** by reducing mechanical wear, and optimizing energy capture through careful bearing selection and design choices; done iteratively through simulation and testing.

Machining Course - George Brown & UofT

Oct 2025

- Operated **fabrication machines** such as a lathe, milling machine, and drill presses, to fabricate various mechanical components with precision; components were inspected to ensure **tolerances** were maintained.
- Analyzed **mechanical drawings** and assembly instructions to **fabricate and assemble** a pneumatic piston and cylinder with **practical tolerances**; assembly success was ensured through testing.
- The success of the assembly resulted in certification needed to use the U of T machine shop.

Team Captain - FIRST Robotics Team 2198

Sep 2022 – Jun 2024

- Led an **interdisciplinary team of 60 students** across mechanical, electrical, and software subteams to design, integrate, and compete **3 robots** at the provincial level, ensuring seamless system-level performance.
- Coordinated cross-team integration by aligning mechanical design, electrical layouts, and control systems, resolving interface conflicts and ensuring compatibility under competition timelines of **3 months**.
- Designed and fabricated robot components using **OnShape** and **SolidWorks**, while enforcing safe operation of manual and CNC **manufacturing equipment**.
- Mentored and onboarded **20+ new members**, establishing standardized workflows, machine shop training, and safety procedures.